



N-FRIENDS PROBLEM:
GENERALIZED FORMULA FOR STEPS COUNT

STATEMENT OF THE PROBLEM

N students (N_C from computer engineering and N_D from data science and engineering) spent the afternoon studying Computational Intelligence. In the evening, they decide to go out for pizza, but the pizzeria can't be reached by walk. One of the students has a tandem bicycle that can transport two people. Find the shortest path to get everyone to the pizzeria.

If there are more computer engineers (C) than data scientist (D) in a group, the evening will be ruined.

- Representation:

(friends at home, friends in pizzeria, * marks position of the tandem)

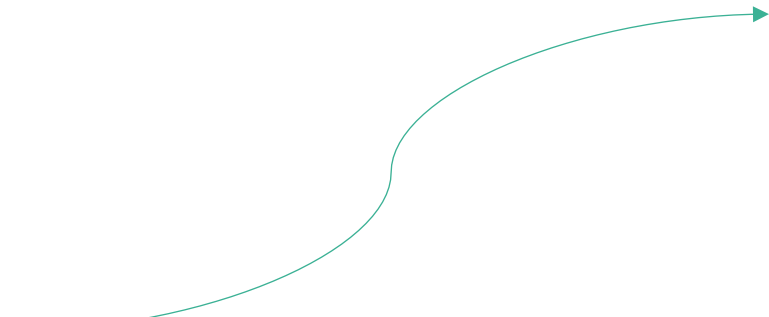
- Constraints:

$$\text{count}(C) \leq \text{count}(D) \text{ or } \text{count}(D) == 0$$

NUMBER OF STEPS

- $N = 2 \rightarrow 1$ step
- $N = 4$ ($N_C = N_D = 2$) $\rightarrow 5$ steps
- $N = 6$ ($N_C = N_D = 3$) $\rightarrow 11$ steps
- $N = 8$ ($N_C = N_D = 4$) $\rightarrow$ no solution

From $N_C = N_D > 3$, no solutions are found.

- 
- 0) CCCDDD*, -
 - 1) CDDD, CC*
 - 2) CCDDD*, C
 - 3) DDD, CCC*
 - 4) CDDD*, CC
 - 5) CD, CCDD*
 - 6) CCDD*, CD
 - 7) CC, CDDD*
 - 8) CCC*, DDD
 - 9) C, CCDDD*
 - 10) CD*, CCDD
 - 11) -, CCCDDD*

THUS, THE
PROBLEM
BECOMES
UNSOLVABLE

... DOES IT?

INEQUAL CASES $N_C \neq N_D$

A. Let's consider $0 \leq N_C < N_D$

From the results in the table, we can deduct the generalized formula:

$$\#steps = \begin{cases} 1, & \text{if } N_C = 0 \text{ and } N_D \leq 2 \\ 3, & \text{if } N_C + N_D = 3 \\ 2(N_C + N_D) - 3 & \end{cases}$$

N_C	N_D	# steps
0	1, 2	1
	3	3
	4	5
	5	7
	...	...
1	2	3
	3	5
	4	7
	...	...
2	3	7
...		

INEQUAL CASES $N_C \neq N_D$

B. What about $0 \leq N_D < N_C$?

Here the solutions are not possible (violating the constraint of $\text{count}(D) \geq \text{count}(C)$ from the initial state) unless $N_D = 0$

N_D	N_C	# steps
0	1	1
	2	1
	3	3
	4	5
	5	7
	...	...

$$\#steps = \begin{cases} 1, & \text{if } N_D = 0 \text{ and } N_C \leq 2 \\ 2N_C - 3, & \text{if } N_D = 0 \end{cases}$$

GENERALIZED FORMULA

$$\#steps = \begin{cases} 1, & 0 < N_C + N_D \leq 2 \\ 3, & N_C + N_D = 3 \text{ and } 0 \leq N_C < N_D \\ 2(N_C + N_D) - 3, & N_C + N_D > 3 \text{ and } 0 \leq N_C < N_D \\ 2N_C - 3, & N_D = 0 \text{ and } N_D < N_C \\ N_D^2 + N_D - 1, & 2 \leq N_C = N_D \leq 3 \\ \text{no solution otherwise} \end{cases}$$

OTHER VARIATIONS TO MAKE THE PROBLEM SOLVABLE WITH ARBITRARY

$$N_C == N_D?$$



UPGRADING TANDEM



Let k be the number of seats on a vehicle.

So far, we have been considering $k = 2$. What if there's a way to fit 3 people in one ride?

Now there's the solution for $N_D = N_C = 4$:

- | | |
|----------------|----------------|
| 0) CCCCDDDD*,- | 5) CD,CCCDDDD* |
| 1) CDDDD,CCC* | 6) CCDD*,CCDD |
| 2) CCDDDD*,CC | 7) C,CCCDDDD* |
| 3) DDDD,CCCC* | 8) CD*,CCCDDDD |
| 4) CDDDD*,CCC | 9) -,CCCCDDDD* |

UPGRADING TANDEM



k	$N_C == N_D$	# steps
3	1	1
	2	3
	3	5
	4	9
	5	11
	≥ 6	No solution

$$\#steps = \begin{cases} 2N_C - 1, & 1 \leq N_C = N_D \leq 3 \\ 2N_C + 1, & 4 \leq N_C = N_D \leq 5 \end{cases}$$

UPGRADING TANDEM

Let's add one more seat!



k	$N_C == N_D$	# steps
4	1	1
	2	1
	3	3
	4	5
	5	7
	6	9

$$\#steps = \begin{cases} 1, & 1 \leq N_C = N_D \leq 2 \\ 2N_C - 3, & N_C = N_D \geq 3 \end{cases}$$

Now for any N the problem is solvable because it can be always a pair of C+D riding a vehicle and moving another pair of C+D, so that groups in both places stay balanced.

CHANGING REPRESENTATION



In the representation used so far, the students on the tandem are counted as a part of a place population whenever they arrive at a pizzeria or at home.

What if the person arriving at one place and proceeding to move to another place should not be counted among the group? Let's say, this person waits outside and doesn't have the opportunity to interact with people inside. Students waiting inside see his arrival through a window and send one person to join him on the ride to another location.

CHANGING REPRESENTATION

Let's take $N = 8$:

- | | |
|--|--------------------|
| 0) CCCCDDDD*, - | 7) CDD, (*C) CCDD |
| 1) CCCDDD, (*C) D #leaving C on tandem | 8) CDD (*C), CCDD |
| 2) CCCDDD (*C), D | 9) CD, (*C) CCDDD |
| 3) CCDDD, (*C) CD | 10) CD (*C), CCDDD |
| 4) CCDDD (*C), CD | 11) D, (*C) CCCDDD |
| 5) CCDD, (*C)CDD | 12) D (*C), CCCDDD |
| 6) CCDD (*C), CDD | 13) -, CCCCDDDD* |

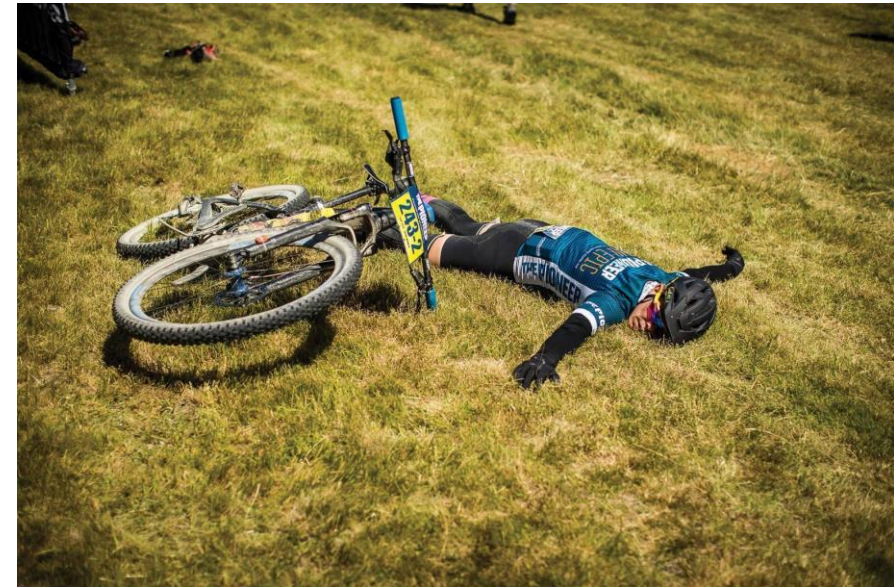
The problem becomes solvable in 13 steps.

CHANGING REPRESENTATION

With the introduced change of the representation, the problems gets trivial.

$N_C == N_D$	# steps
1	1
2	5
3	9
4	13
5	17
6	21

$$\text{\#steps} = 4N_C - 3, \quad N_C = N_D \geq 1$$



The Computer Engineering student after riding everyone to pizzeria

THANK YOU FOR ATTENTION!

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